Data Cleansing & Quality Audit Log

Project: E-Commerce Customer Retention Intelligence
Scope: Registered Customer Cohort Analysis & RFM Segmentation
Status: Verified & Reconciled

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1. Executive Summary & Data Integrity Rules

This audit log documents the data cleaning, transformation, and quality assurance processes applied to the raw e-commerce transaction dataset prior to semantic modeling in Power BI.

To satisfy the **Business Requirements Document (BRD)**, the core data processing strategy isolates **active registered customer behavior**, explicitly separating non-qualifying and guest checkout transactions.

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2. Cleansing Steps & Filtering Logic

Step 1: Order Status Filtering

- **Rule:** Filter strictly for `OrderStatus = "Completed"`.
- **Rationale:** Excludes cancelled, pending, failed, or refunded transactions to ensure monetary calculations reflect true realized revenue.

Step 2: Customer Identity Isolation (Guest Checkout Exclusion)

- **Rule:** Exclude rows where `CustomerID` is null, blank, or anonymous (`CustomerID is not blank`).
- **Impact:**
  - **Guest Checkouts Excluded:** 1,407 orders representing **$101,140.00** in net sales.
  - **Registered Cohort Retained: 9,607 unique registered customers** generating **$3,108,791.35** in completed net sales across **18,214 completed orders**.

Step 3: Date Table Engineering (`dim_date`)

- **Rule:** Built a dedicated, continuous calendar dimension table (`dim_date`) spanning from the minimum transaction date (`2023-01-01`) to the maximum transaction date (`2025-12-31`).
- **Attributes Generated:** `DateKey`, `Year`, `Quarter`, `Month`, `MonthName`, `DayOfWeek`, `FiscalYear`, `IsWeekend`.
- **Relationship:** Joined `dim_date[Date]` (1) to `fact_orders[OrderDate]` (*) to enable time-intelligence calculations, cohort analysis, and period-over-period trend comparisons.

Step 4: Granularity & Order-Grain Verification

- **Rule:** Enforce order frequency at the distinct order grain: `DISTINCTCOUNT(fact_orders[OrderID])`.
- **Rationale:** Prevents multi-item line orders from artificially inflating customer purchase frequency scores.

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3. Reconciled Baseline Metrics

Metric	Raw Dataset	Cleansed Registered Cohort	Variance / Exclusions
Total Customers	11,014	**9,607**	-1,407 (Guest Checkouts)
Completed Orders	19,621	**18,214**	-1,407 (Guest Orders)
Total Net Revenue	$3,209,931.35	**$3,108,791.35**	-$101,140.00 (Guest Sales)
Avg Spend / Registered User	N/A	**$323.60**	Baseline established

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4. Sign-Off & Verification

- **Data Quality Passed:** All customer IDs are non-null and mapped to stable keys.
- **RFM Scoring Engine Ready:** Scores (1–5) for Recency, Frequency, and Monetary value calculated across all 9,607 registered accounts.